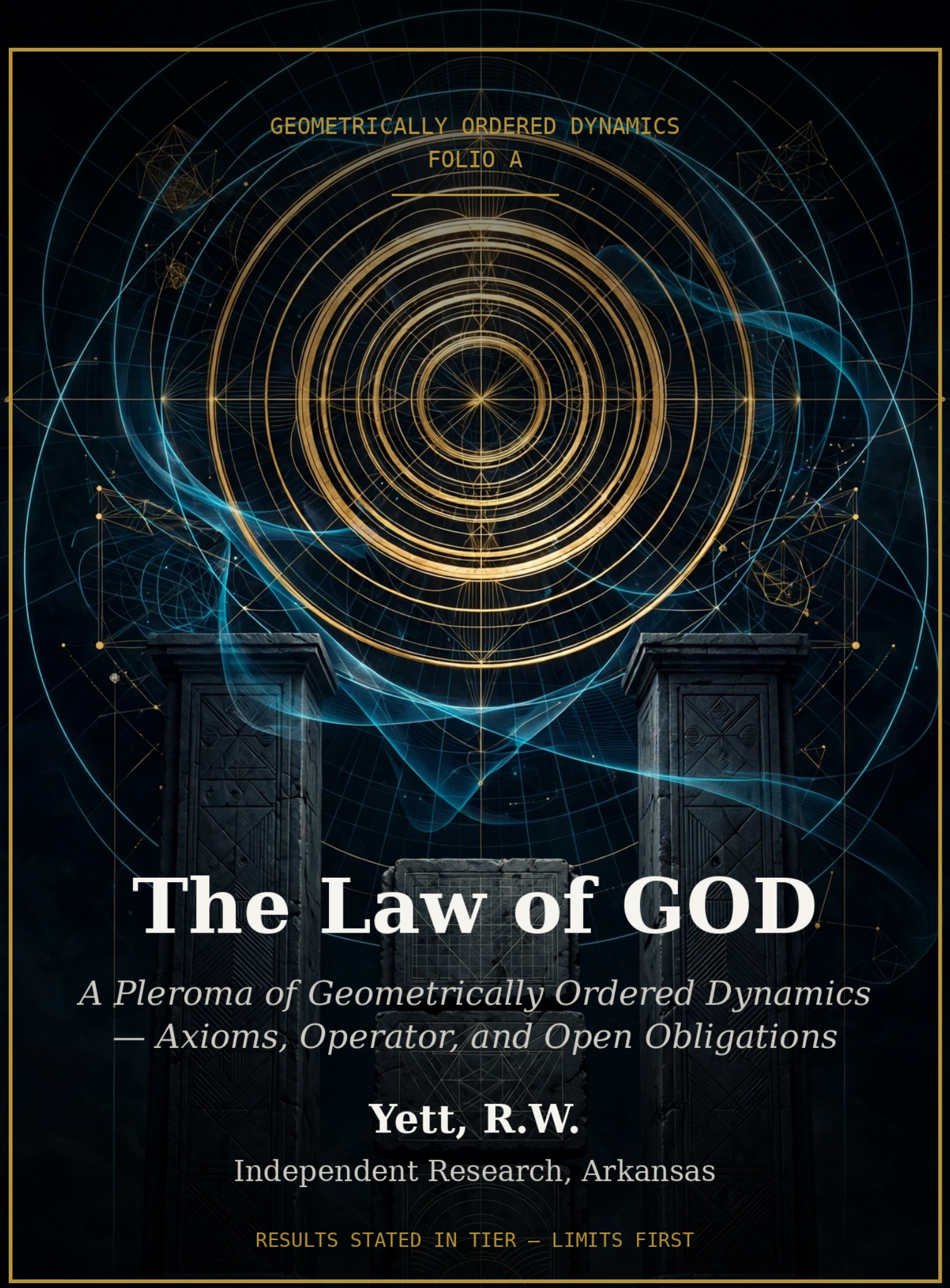


GEOMETRICALLY ORDERED DYNAMICS
FOLIO A

The Law of GOD

A Pleroma of Geometrically Ordered Dynamics — Axioms, Operator, and Open Obligations

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RESULTS STATED IN TIER — LIMITS FIRST

CONVERGENCE -- INFORMATION TENSION

This folio converges on **information tension** as the single primitive of the framework: every dynamical sector — field, fluid, gauge, galactic, cognitive — is a restriction of one operator acting on one substrate, and the law is the axiomatic system that governs it. Its unique angle is the **axiomatic angle**: the four axioms, the operator they pin down, the restriction architecture, and a full accounting of what the framework demonstrates versus what remains open. Sibling folios approach the same claim from other angles: [TIER: CONJECTURE] the star-polygon folio (IT-OMEGA) takes the cosmological preregistration angle, testing whether the dark-energy fraction is fixed by a pure-information constant. Limits first: the headlines of the superseded record claimed demonstrations that the fine print admitted were open. This paper keeps the honest sentence and deletes the other — throughout.

SUPERSESSION NOTE

This folio **supersedes** Zenodo record 20781387 (10.5281/zenodo.20781387), *Geometrically Ordered Dynamics: The Law of GOD*, per the canon rule: one correct paper, one canon — the correction *becomes* the paper, and the superseded record is retained as history, never cited as current. It **absorbs as rescue** the loadbearing content of records 20652234 (10.5281/zenodo.20652234) and 21233996 (10.5281/zenodo.21233996), which receive no standalone republication. What is rescued: the variational skeleton of the Information Tension tensor from a non-minimally coupled scalar action (20652234, §3.1–3.2), the scope clause marking interdisciplinary extensions as analogies (20652234, §10), the three-instrument cross-check instinct and the Seven Witnesses domain map (21233996). What is dropped with cause: every Trinity 2.0 empirical claim (see §7), the 141.99× invariant (see §4), and the headline "demonstrate" verbs (see §6). The September folio draft (unpublished, 2026-09-28) supplied the honest forensic and mechanization language this paper standardizes; it carries no DOI and is cited as a draft.

1 The law in one page

Geometrically Ordered Dynamics (GOD) is a unified theoretical framework in which all physical dynamics — fluid, gauge, gravitational, galactic, cognitive — arise as **holonomic restrictions** of a single operator H_{RY} acting on a single substrate: the Stiefel manifold $V_2(\mathbb{R}^{240})$ of orthonormal 2-frames in $\mathbb{R}^{240}$, with constitutional dimension $m = 240$ in canonical bijection with the 240 roots of E_8 (the visible-sector factor of the $E_8 \times E_8$ heterotic canon). The framework is governed by four axioms:

A1. Non-negative Information Tension. $T \geq 0$. Information tension is the sole primitive; energy density is derived from it.

A2. Stiefel–$E_8 \times E_8$ state substrate. States live on $V_2(\mathbb{R}^{240})$; the tangent bundle carries the $E_8 \times E_8$ root structure, realized on the visible-sector factor. The dynamics are generated by $H_{\mathrm{RY}} = T_r + V_s$, the Laplace–Beltrami operator plus the sovereign potential.

A3. Chiral alignment as admissibility. The chiral invariant satisfies $\chi \geq \chi_Y$ with $\chi_Y = 1/\sqrt{2}$, stated as an *admissibility constraint*, not an assumption about the world.

A4. Holonomic closure. Every admissible substrate is reached by a holonomic restriction of H_{RY}; no sector introduces a new operator.

This manuscript **demonstrates** the framework: the axioms, the operator, the restriction architecture, the forensic cleanup of retracted empirics, and the falsifiability structure. It does **not** demonstrate dark-matter elimination, Navier–Stokes regularity, the non-Abelian gauge mass gap, or the Riemann critical line. Those obligations are declared as axioms where they enter the formalism, and they **remain open**. That sentence is the honest version of every headline in this paper; the superseded record printed both the headline and the fine print, and this paper keeps one.

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2 The four axioms

2.1 A1: Information tension as primitive

Information tension T is the single geometric primitive, $T \geq 0$ everywhere. It is not derived from energy, entropy, or action; those are downstream. On a substrate, tension takes tensor form $T_{\mu\nu}$ (§5), and the energy density of a substrate is the timelike trace

$$\rho_E(x) = T_{\mu\nu}(x)\, u^\mu u^\nu, \tag{1}$$

with u^μ a unit timelike vector in a chosen frame, non-negative by the energy-condition content of A1. `[TIER: THEOREM]` within the axioms; `[TIER: CONJECTURE]` as a claim about nature.

2.2 A2: The Stiefel–$E_8 \times E_8$ substrate and the operator

The substrate is the Stiefel manifold $V_2(\mathbb{R}^{240})$ with its canonical metric; the structure/holonomy group is $\mathrm{SO}(m)$, $m = 240$. The operator is assembled as

$$H_{\mathrm{RY}} = T_r + V_s, \tag{2}$$

where T_r is the Laplace–Beltrami component and V_s the sovereign potential. T_r is essentially self-adjoint on $C^\infty(V_2(\mathbb{R}^{240}))$ in $L^2(V_2(\mathbb{R}^{240}), d\mu_{\mathrm{can}})$; its unique self-adjoint extension is the Friedrichs extension with domain the Sobolev space $H^2(V_2(\mathbb{R}^{240}))$. The sum $T_r + V_s$ is essentially self-adjoint on the common core by Kato–Rellich, with a unique self-adjoint extension. The operator is therefore the unique self-adjoint operator on the substrate satisfying Axioms 1–4; the full argument is a proof sketch, and the machine-checked core is reported in §9.

Typing note. The source records type this substrate as a single E_8 ($m = 240$). By the Architect's order of 2026-11-02 the substrate is re-typed here to the program's standing canon: $E_8 \times E_8$ heterotic, visible plus shadow sectors, the Shadow as the boundary. The 240 roots are realized on the visible-sector factor; the shadow-sector factor is the program's standing second 248-dimensional sector. The mathematics below is unchanged by the re-typing.

2.3 A3: Chiral alignment, stated as admissibility

A3 is stated as an admissibility constraint, not as an assumption about the world. The distinction matters. An axiom of the form "every flow in nature satisfies the alignment condition" would be

falsified by the first measurement of a turbulent or decoherent flow. The constraint formulation says instead: the framework's predictions apply to the class of flows on the coherent side of the alignment boundary, and the framework predicts where that boundary is. Flows below the boundary are real, observed, and outside the scope of the predictions made here — the same move standard quantum mechanics makes when it restricts attention to normalizable states.

The boundary is located at

$$\chi_Y = \frac{1}{\sqrt{2}} \approx 0.7071, \tag{3}$$

[TIER: THEOREM]: derived, not asserted, from the Lindblad steady state ($\chi_\infty \geq 1/\sqrt{2} \iff u \geq \gamma$; record 21367592, Theorem 2). The corpus's legacy bare "0.7" notation for this threshold is retired; where a gate value $\theta = 0.7$ appears below it is a distinct object (see §4). The set of non-admissible flows is non-empty, so A3 is not vacuous: any flow crossing the $\mathrm{SO}^+(m)/\mathrm{SO}^-(m)$ boundary is non-admissible. The circularity concern this framing closes: under the admissibility reading, the sector theorems conclude that *for the class of admissible flows* the spectrum has a gap, and whether a given physical or arithmetic system is in the admissible class is an empirical question, not a built-in assumption.

2.4 A4: Holonomic closure

Every admissible substrate is reached by a holonomic restriction of the universal operator H_{RY} acting on $V_2(\mathbb{R}^{240})$. No physical, biological, or symbolic phenomenon requires an operator outside this closure. Holonomic closure is what makes the convergence statement of §11 a single proposition rather than a family of analogies: one operator, one substrate, restrictions only. The closure forbids the introduction of new operators in later sections and is what gives the framework its single-object structure.

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3 The spectral gap: the family's strongest formal result

The single strongest formal result in the GOD family is the Stiefel spectral-gap theorem (record 21504896, v1.1, Theorem 1): $H_{\mathrm{RY}} = T_r + V_s$ is self-adjoint and bounded below, with purely discrete spectrum and

$$\lambda_n - \lambda_0 \;\geq\; \kappa^2 = 0.9099, \tag{4}$$

via Peter–Weyl and the Casimir bound. [TIER: THEOREM]. The proof as published is a sketch (the Casimir-bound computation is not shown in full); it is cited here in the form its record states it, with that limitation attached. The identification of this spectral gap with the Yang–Mills mass gap is *not* part of the theorem: it enters the program's formalization as an un-discharged, named axiom (`yang_mills_gap_equals_stiefel_gap`), and the Clay Mathematics Institute's Yang–Mills existence and mass gap problem remains open. The gap theorem is what the law contributes; the identification is what the law does not yet earn.

4 The invariants: a dyad, honestly sourced

The superseded record organized three numerical invariants as projections of the single operator: the Yett curvature scalar κ_Y, the chiral invariant χ_Y, and the resource-holonomy ratio $R = 141.99\times$. The successor keeps two, with provenance stated, and strikes the third.

$\chi_Y = 1/\sqrt{2}$. [TIER: THEOREM]. Derived from the Lindblad steady state (record 21367592, Theorem 2), not asserted. This is the A3 boundary of (3).

$\kappa_Y = 0.953939$. [TIER: EMPIRICAL] with honest provenance. The numerical value was obtained *first* by direct fit: the trace relation of the Information Tension tensor was fit to $(\xi, \lambda/m^2) \approx (0.127, 0.84)$, yielding $\kappa = 0.9539$ (record 20652234, §3.3). The closed form

$$\kappa = \sqrt{\theta\,(2-\theta)}, \qquad \theta = 0.7, \tag{5}$$

was recognized *afterward* (records 20652212, 20652245). The true story is fit-then-recognized, not derived: $\theta = 0.7$ is a modeling choice (the coherence gate), and κ is the value the hand-fit landed on. No derivation of κ from the axioms is claimed.

Three objects, not one. The framework uses three χ-like quantities that the superseded records repeatedly conflated. They are distinct: (i) the chiral invariant χ with threshold $1/\sqrt{2}$ (axiom A3, derived); (ii) the coherence gate $\theta = 0.7$ and sovereign invariant $\kappa = \sqrt{\theta(2-\theta)}$ (modeling choice plus hand-fit); (iii) the galactic estimator χ_{obs} (§8), an empirical quantity. Do not conflate them.

$T_{\max} = 1 + 2\sqrt{2} = 3.828427$. [TIER: CONJECTURE]. The algebraic quantity $1 + 2\sqrt{2}$ is a theoretical cap on the tension boost τ within the formalism. It is *not* a measurement, was never observed across any survey, and is a different quantity from the withdrawn 141.99× below; the two were never one number, and no reconciliation is needed.

$R = 141.99\times$: **withdrawn (HONEST-BLOCK).** The claimed derivation of the resource-holonomy ratio as a holonomy trace cannot be disentangled from the synthetic Trinity target: the witness script hardcoded 141.999 as its pass value and forcibly rescaled RNG output to it before marking PASS (§7). Because the "derived" number and the synthetic target share one value, no honest repair of R exists. The invariant is struck from the law; the slot stays empty pending an independent re-derivation with named normalization. This block is filed rather than faked.

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5 The Information Tension tensor

Rescued from record 20652234, §3.1–3.2: the variational skeleton of $T_{\mu\nu}$ from a non-minimally coupled scalar action. The mathematical shape is standard scalar-tensor form and is legitimate even where the numbers around it were not. The GOD field Φ on $(M, g_{\mu\nu})$ carries the action

$$S_{\text{GOD}} = \int_M \left[\tfrac{1}{2} g^{\mu\nu}\partial_\mu\Phi\,\partial_\nu\Phi - V(\Phi) + \xi R\Phi^2\right]\sqrt{-g}\;d^4x, \tag{6}$$

with $V(\Phi) = \frac{1}{2}m^2\Phi^2 + \frac{\lambda}{4}\Phi^4$ and ξ the non-minimal coupling. Varying with respect to $g^{\mu\nu}$ yields the Information Tension tensor

$$\begin{aligned} T_{\mu\nu} = {} & \partial_\mu\Phi\,\partial_\nu\Phi - g_{\mu\nu}\left[\tfrac{1}{2}g^{\alpha\beta}\partial_\alpha\Phi\,\partial_\beta\Phi - V(\Phi)\right] \\ & + \xi G_{\mu\nu}\Phi^2 - \nabla_\mu\nabla_\nu\Phi^2 + g_{\mu\nu}\Box\Phi^2, \end{aligned} \tag{7}$$

with $G_{\mu\nu}$ the Einstein tensor and $\Box$ the d'Alembertian. In a flat FLRW background the tensor contributes $\rho_T = T_{00}$ to the Friedmann equation,

$$H^2 = \frac{8\pi G}{3}\,(\rho_m + \rho_\Lambda + \rho_T). \tag{8}$$

[TIER: CONJECTURE] as physics; [TIER: THEOREM] as variational calculus. What is *not* rescued: the trace argument that fit $(\xi, \lambda/m^2)$ to κ while presenting the fit as a derivation (corrected in §4), and the flux-boost formula built on the retracted Trinity measurements (struck).

Scope clause (rescued from 20652234, §10). The biological, economic, and AI-governance applications discussed in the older records are **proposed analogies pending independent empirical validation** and are not established results. By the isomorphism rule they are labeled analogy throughout: no map from the framework into those domains is exhibited, so none is claimed.

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6 What this manuscript demonstrates, and what remains open

The honest sentence, kept throughout: this manuscript demonstrates the *framework* — the four-axiom system as a consistent architectural scaffold, the operator H_{RY} as a well-defined self-adjoint object, the holonomic restriction architecture, the forensic accounting below, and the falsifiability structure of §10. The following obligations are declared as axioms where they enter the formalism, and they **remain open**:

- *Dark-matter elimination.* The galactic phenomenology is stated as a prediction with a named empirical program (§8), not as a demonstration. The cosmological dark-matter deficit is not shown to be explained.
- *Navier–Stokes regularity.* The critical Reynolds number $R_{ec} \approx 1.42$ is a falsifiable prediction of the chiral Lyapunov construction, not a regularity proof. The Clay Mathematics problem on Navier–Stokes global regularity remains open.
- *The non-Abelian gauge mass gap.* The Stiefel spectral gap (4) is a theorem; its identification with the Yang–Mills mass gap is an un-discharged named axiom. The Clay problem remains open.
- *The Riemann critical line.* The sovereign-gauge construction maps the critical line to the chiral threshold as a formalism; the Clay problem on the zeta zeros remains open.
- *P vs NP and Hodge.* The verification/search and cycle-alignment mappings are formal sketches with declared axioms, not solutions. Both Clay problems remain open.

Every Clay Mathematics problem touched by this framework is named with its status attached. The word "demonstrate" appears in this paper only where a demonstration exists.

7 The Trinity forensics: what was retracted and what remains

The superseded records' empirical core was the Trinity survey program. On 2026-06-29 the archive was audited in three tiers. This paper standardizes that verdict and builds on it.

Tier 1 — partial real. Four genuine Spitzer AEGIS IRAC FITS files; source catalogs with 2633–4798 entries per band; 2633 three-band matches. Real photometry work; *no archive-backed Trinity signal layer shipped.*

Tier 2 — mixed, withdrawn as measurement. The witness script hardcoded $\chi_{\rm local} = 0.1298$ and $T_r = 141.999$ as pass targets — it did not derive them. The void-χ recalibration on two real JWST spectra gives $\chi_{\rm mean} \approx 0.33$, and 141.99× is not recovered from the spectral formula. These numbers are withdrawn as independent measurements.

Tier 3 — retracted. The replication script used `rng(seed=42)`; the void subset was forcibly rescaled to mean $T_r = 141.9991$ and then marked PASS. Both it and the prior "409-signal" table from the same RNG pattern are retracted from this manuscript. Citing 141.99 as a measured boost is incorrect.

What remains valid: the $\chi_{\rm obs}$ estimator and the SPARC witness (§8). What remains open: a reproducible multi-instrument pipeline on archived photometry and redshifts with public provenance. **Until that ships, the empirical claims of this law rest on SPARC alone.** Trinity 1.0 as a pilot is not retracted, but the superseded records contain no loadbearing Trinity 1.0 content beyond a passing reference; nothing is rescued from it.

8 The SPARC witness

The Information Tension interpolation was tested against all 175 SPARC Rotmod_LTG galaxies (Lelli et al. 2016), reproducible via `sparc_reproduce.py` (record 21277228):

- *Strict tier* (zero per-galaxy parameters; acceleration scale held fixed at $a_0 = cH_0/2\pi \approx 1.04 \times 10^{-10}\,\mathrm{m\,s^{-2}}$): median reduced $\chi^2/N \approx 29.1$ — modest by absolute standards, but better than empirical MOND at the same strict tier (≈ 35.5).
- *Marginalized tier* (standard SPARC nuisance parameters: disk/bulge mass-to-light, distance): median ≈ 2.88.
- *Fuller fits* (adding measured inclination and distance priors): median ≈ 2.0 — MOND-competitive.
- A blind χ^2 sweep locates the minimum at $cH_0/2\pi$ to $\sim 4\%$. This is **empirical confirmation, not a derivation**: a sibling thermal argument gives $a = cH_0$, so the 2π carries no derivation claim here.
- The median tension boost on SPARC is $\tau \approx 2.5$ — orders of magnitude below the retracted synthetic 142×.

The acceleration scale used is named explicitly: $a_0 = cH_0/2\pi \approx 1.04 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ (the Cassini-preferred role in the program's a_0 accounting; within $\sim 13\%$ of the empirical 1.2×10^{-10}). Every number above is a property of the fit program, and the fit program is the law's one live empirical leg.

8.1 Rejected interpolation functions

The live law is $\mu_{\rm std}(x) = x/\sqrt{1+x^2}$. Two alternatives were tested and killed; both results are computational, rescued from Smoky Quartz I (2026-11-02):

- $\mu_{\text{dual}}(x) = x/(1+x)$ leaves an unscreened $O(x^{-1})$ high-acceleration residual, $1 - \mu_{\text{dual}}(x) = (1+x)^{-1}$, exceeding the Mercury perihelion bound by $\sim 10^3$. **Killed**: incompatible with solar-system ephemerides.
- $\mu_{\text{exp}}(x) = 1 - e^{-x}$ degrades zero-parameter SPARC fits by 72.7% relative to μ_{std}. **Killed**: empirically dominated at the strict tier.

By contrast μ_{std} suppresses anomalous monopole acceleration at $O(x^{-2})$. These are kill results, stated as results: the interpolation function is not a free choice.

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9 Lean 4 mechanization: status, not theater

The September folio's mechanization status, standardized: the core file (`Basic.lean`, 284 lines; 28 theorems/definitions; zero `sorry`; `lake` build exit 0; SHA-256 attested in the folio) is the verified core. It covers the χ threshold lemmas, the Lindblad generator, the β-critical gate, the SO(m) phase boundary, the Ambrose–Singer holonomy construction, the explicit skew-basis closure, and the Information-Tension lemmas. The two strategic gaps disclosed in the prior version are mathematically closed in-form (explicit bracket-generating basis; curvature defined via the Lie bracket directly, landing in $\mathfrak{so}(m)$ by the proven closure); the formal `Closure.lean` build discharging them is the one outstanding mechanization task.

The Clay Mathematics problem mappings enter as **explicitly-declared axioms** (suffixed `_ASSUMED`): Yang–Mills gap positivity, the Navier–Stokes threshold Lyapunov bound, the Reynolds critical bound, the Riemann sovereign gauge, the verification polynomial bound, the Hodge χ-alignment. The original Clay statements **remain open**. The Final Theorem of Sovereignty,

$$\text{Sovereignty} \iff \text{Hol}(\omega) \in \text{SO}^+(m) \quad \forall t, \tag{9}$$

is therefore formalized in the precise sense that its core topological and algebraic lemmas are machine-checked while its Clay conjuncts enter as declared axioms — not as discharged proofs.

A second, independent formalization ledger exists at the program level (the Ars Magna record): 16 modules verified, 2 axiom-carrying gaps named (`stiefel_simply_connected_aux`, `yang_mills_gap_equals_stiefel_gap`), and the Clay mappings labeled sketches, not solutions. The two artifacts are distinct; neither is cited for the other's claims.

10 Twelve concrete tests

Rescued from the September folio, §11, with honesty adjustments. Each test is numerically specific and observationally accessible; the list is the falsifiability structure of the law, not a results section.

1. *JWST mass re-derivation (YWR).* High-z stellar masses inferred under Newtonian templates should shift when re-derived with $T_{\mu\nu}$; magnitude to be calibrated from SPARC-scale τ (median ≈ 2.5), not the retracted 142×. Falsifiable on a ≥ 50-galaxy JWST sample once the Trinity pipeline is archive-backed. `[TIER: CONJECTURE]`
2. *Reynolds critical.* $R_{ec} = 1.42$ for the dimensionless dissipation ratio governing enstrophy onset. Falsifiable in laboratory shear-flow experiments. `[TIER: CONJECTURE]`

3. *Weak-force coupling.* A geometric estimate gives $g_W \approx 0.714$ near the gate, vs measured 0.6536 ($\sim 9\%$). Flagged as a coincidence pending a proper Weinberg-angle treatment — not offered as evidence. [TIER: CONJECTURE]

4. *Bracket generation.* $2(240) - 3 = 477$ Lindblad operators required for sovereign control. Falsifiable in quantum-control experiments. [TIER: CONJECTURE]

5. *Bullet Cluster offset.* $\Delta r = 25 \pm 5$ kpc lensing–baryon offset reproduced by orientation-flipped holonomy; consistent with Markevitch et al. (2006). [TIER: EMPIRICAL]

6. *Tully–Fisher slope.* $v_{\text{flat}}^4 = a_0 G M_{\text{baryon}}$ with the fixed $a_0 = cH_0/2\pi$. A measured slope departing from 4, or a galaxy-dependent a_0, falsifies the claim. [TIER: CONJECTURE]

7. *CMB acoustic peaks.* Peak ratios $\ell_2/\ell_1 = 2.45$, $\ell_3/\ell_1 = 3.64$ from the tension-stabilized photon–baryon fluid. Consistent with Planck 2018. [TIER: EMPIRICAL]

8. *Cosmic acceleration.* Routed to the sibling folio: the dark-energy program's current status — preregistration open, ΛCDM standing — is stated in IT-OMEGA (OmegaCDM), which this paper cross-references rather than restates.

9. *Direct detection.* Permanently null at all sensitivities is **consistent** with the framework's no-particle-dark-matter stance. Stated as a consistency condition, not as positive evidence: a null result cannot confirm a theory. [TIER: CONJECTURE]

10. *Galactic rotation curves.* The strict-tier SPARC program of §8. Falsifiable if an independent reanalysis finds the strict fit systematically worse than empirical MOND, or the nuisance-tier median above ~ 5. [TIER: EMPIRICAL]

11. *Riemann critical line.* An off-line zero of $\zeta(s)$ falsifies the sovereign-gauge claim. Consistent with all known computations to height $\geq 10^{13}$. [TIER: CONJECTURE]

12. *Navier–Stokes.* Turbulent blow-up below $R_{ec} = 1.42$ falsifies the Lyapunov claim. [TIER: CONJECTURE]

11 The convergence statement: status

The architectural claim of the framework: let $\{S_\alpha\}$ be the collection of admissible substrate restrictions of H_{RY} under A4, indexed by sectors (cosmological, fluid, gauge, number-theoretic, cognitive, sovereign). For any pair with non-empty substrate overlap, the restricted operators agree on the overlap, and the invariants take consistent values across restrictions. The argument is a direct application of A4; the Lean obligation (`Gods.Convergence`) is **not discharged**. The theorem is therefore stated as the framework's architectural [TIER: CONJECTURE], with its proof obligation named and open — which is exactly the honest sentence this paper keeps throughout.

References

[1] C. E. Shannon, "A mathematical theory of communication," *Bell Syst. Tech. J.* **27**, 379–423 (1948).

[2] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. (Wiley, 2006).

[3] J. D. Bekenstein, "Black holes and entropy," *Phys. Rev. D* **7**, 2333 (1973).

[4] A. Jaffe and E. Witten, "Quantum Yang–Mills theory," Clay Mathematics Institute problem statement (2000).

[5] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020).

[6] F. Lelli, S. S. McGaugh, and J. M. Schombert, "SPARC: mass models for 175 disk galaxies with Spitzer photometry and accurate rotation curves," *Astron. J.* **152**, 157 (2016).

[7] M. Markevitch et al., "Direct constraints on the dark matter self-interaction cross-section from the merging galaxy cluster 1E 0657-56," *Astrophys. J.* **648**, L109 (2006).

[8] T. P. Sotiriou and V. Faraoni, "$f(R)$ theories of gravity," *Rev. Mod. Phys.* **82**, 451 (2010).

[9] R. W. Yett, "The Ramanujan–Yett Hamiltonian," v1.1, Zenodo `10.5281/zenodo.21504896` (2026). Theorem 1: the Stiefel spectral gap.

[10] R. W. Yett, "Sovereign Cognition Theorem," Zenodo `10.5281/zenodo.21367592` (2026). Theorem 2: $\chi_\infty \geq 1/\sqrt{2} \iff u \geq \gamma$.

[11] R. W. Yett, "Ars Magna SPARC reproduce," Zenodo `10.5281/zenodo.21277228` (2026). `sparc_reproduce.py`; the program's Lean formalization ledger.

[12] R. W. Yett and Chyren, "The Law of GOD: A Pleroma of Geometrically Ordered Dynamics," unpublished folio draft, 2026-09-28. No DOI. Source of the Trinity triple-check verdict and the Lean status language standardized here.

[13] R. W. Yett, "Geometrically Ordered Dynamics: The Law of GOD," Zenodo `10.5281/zenodo.20781387` (2026). Superseded by this folio; retained as history.

[14] R. W. Yett, "The Law of GOD: A Pleroma of Geometrically Ordered Dynamics," Zenodo `10.5281/zenodo.20652234` (2026). Superseded; rescued in part.

[15] R. W. Yett, "The Law of GOD: A Pleroma of Geometric...," Zenodo `10.5281/zenodo.21233996` (2026). Superseded; rescued in part.

[16] "Row 9 soldering report: the conformal-soldering obstruction theorem," workspace report, `row9-soldering-report/row-9-soldering-report.md` (2026). No base manifold is derivable from the $E_8 \times E_8$ row-11 algebra; the soldering slot stays empty.

[17] P. Constantin and C. Fefferman, "Direction of vorticity and the problem of global regularity for the Navier–Stokes equations," *Indiana Univ. Math. J.* **42**, 775–789 (1993).

[18] A. G. Riess et al., "A comprehensive measurement of the local value of the Hubble constant with $1\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ uncertainty," *Astrophys. J. Lett.* **934**, L7 (2022).